Spotify

Data & Insights Team

Our Team
Information
Dataset Structure
Business Inquiries
Analysis
Future Considerations
Thank you

Our Team

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Artist

Simone Cobuccio

Dj

Giancarlo Ferrara

Vocalist

Davide Tissino

Instrumentalist

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Information

Stakeholder: “Spotify Personalisation & Discovery Team”

Objective: Analyze the content discovery algorithm

Problems: Uncover which audio features and genres are most strongly associated with high track popularity

Goal: Identify interpretable audio features and content-based recommendation strategies

Play
Follow

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Dataset Structure

Track Metadata

- Track_ID
- Album_name, Artist_name
- Popularity
- Duration_ms
- Explicit
- Track_genre

Audio Features

- Danceability
- Acousticness
- Energy
- Instrumentalness
- Key
- Liveness
- Loudness
- Valence
- Mode
- Tempo
- Speechiness

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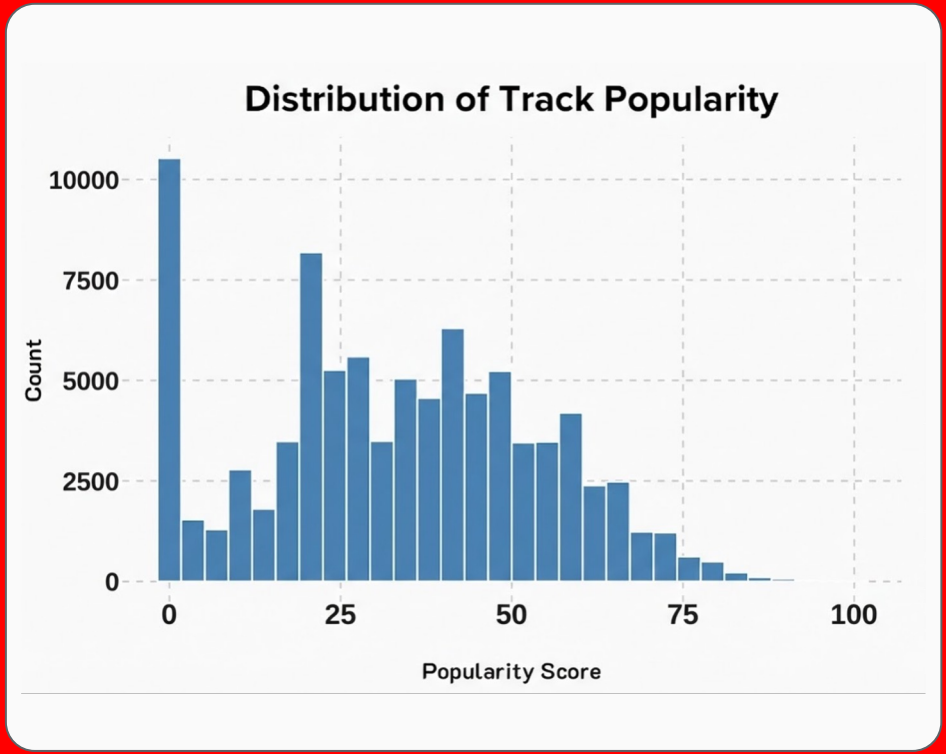
Spotify

Business Inquiries

#	Title	Meaning
1	**Spotify Winner-Takes-All Economy**	The Landscape
2	**Standard Production Rules & Market Gaps**	The Interactions
3	**How Algorithms Cluster Music**	The Reality Check
4	**Predicting Hit Valuation through Sonic Math**	The Ultimate Test

What do you want to analyse?
D&I

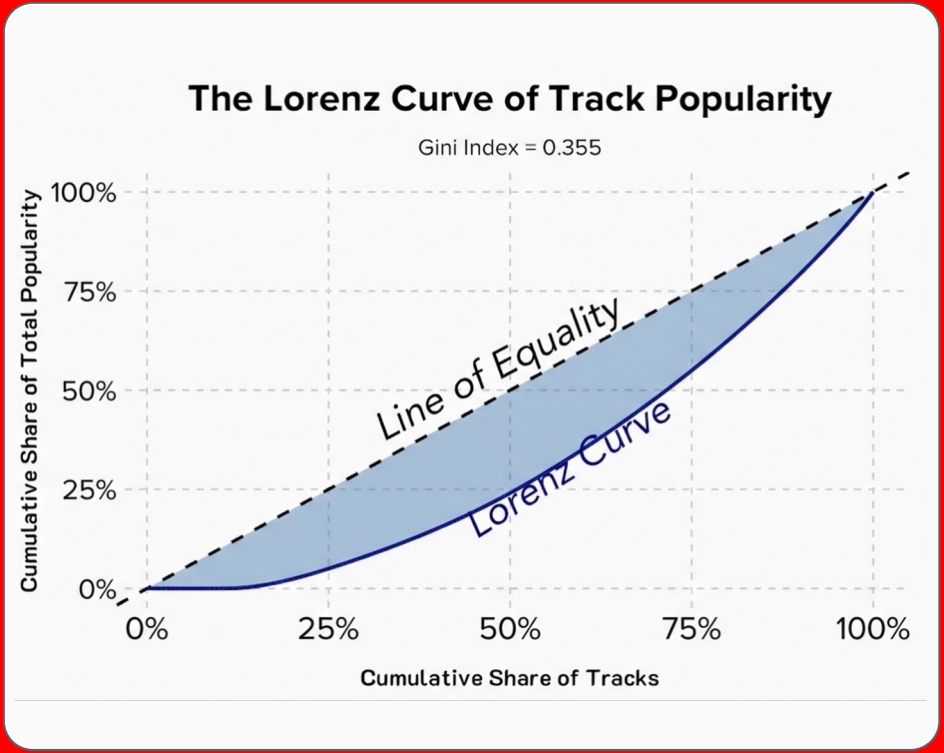
Spotify winner-takes-all economy



Histogram of Track Popularity
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

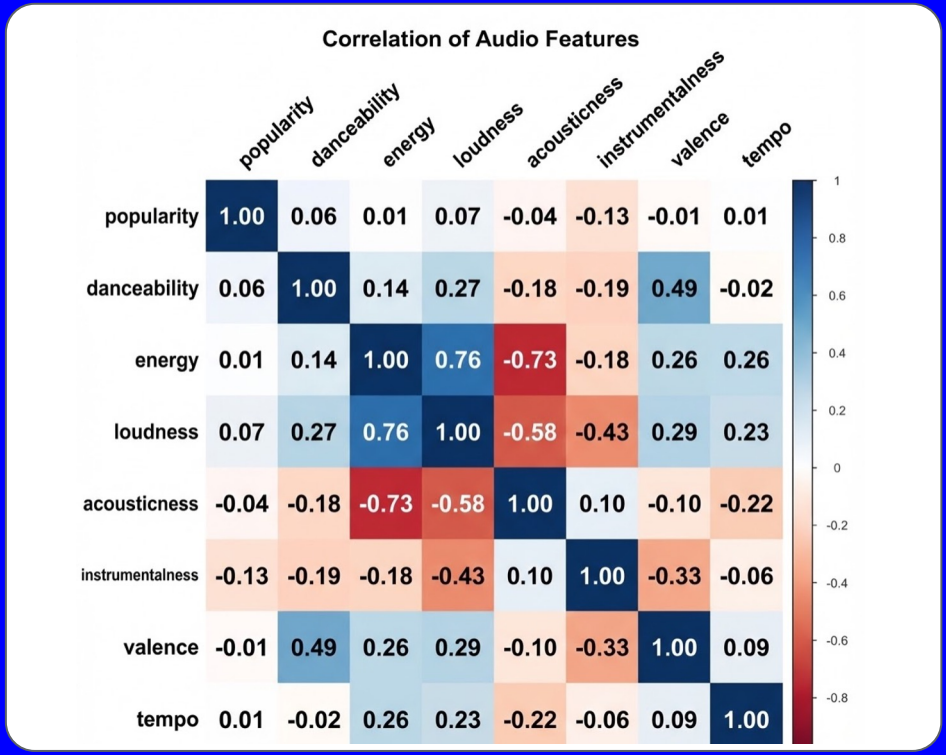
Spotify winner-takes-all economy



Lorenz Curve of Track Popularity
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

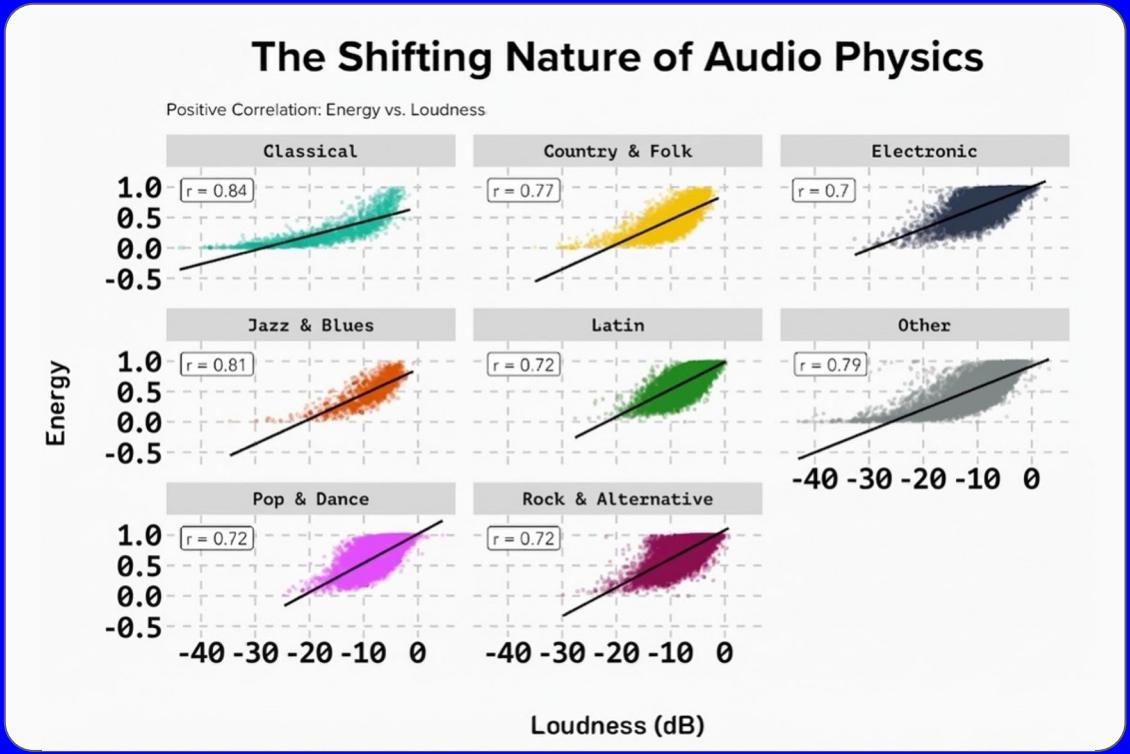
Standard Production rules & Market gaps



Correlation of Audio Features
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

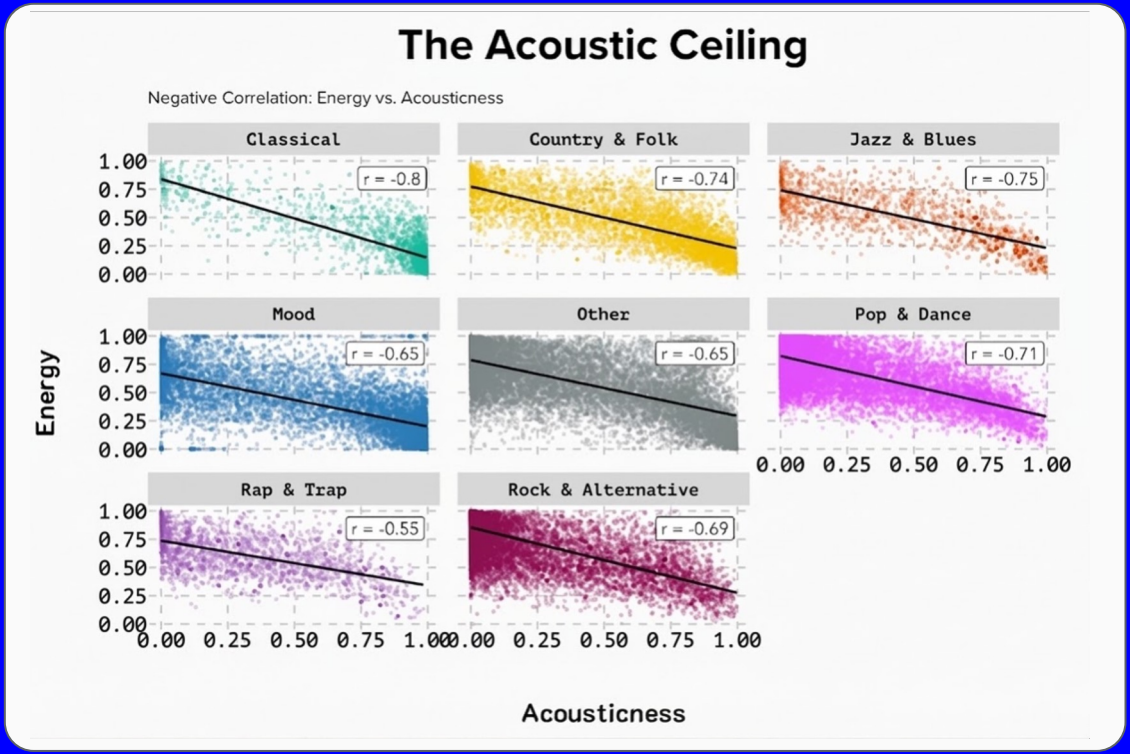
Standard Production rules & Market gaps



Positive Correlation
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

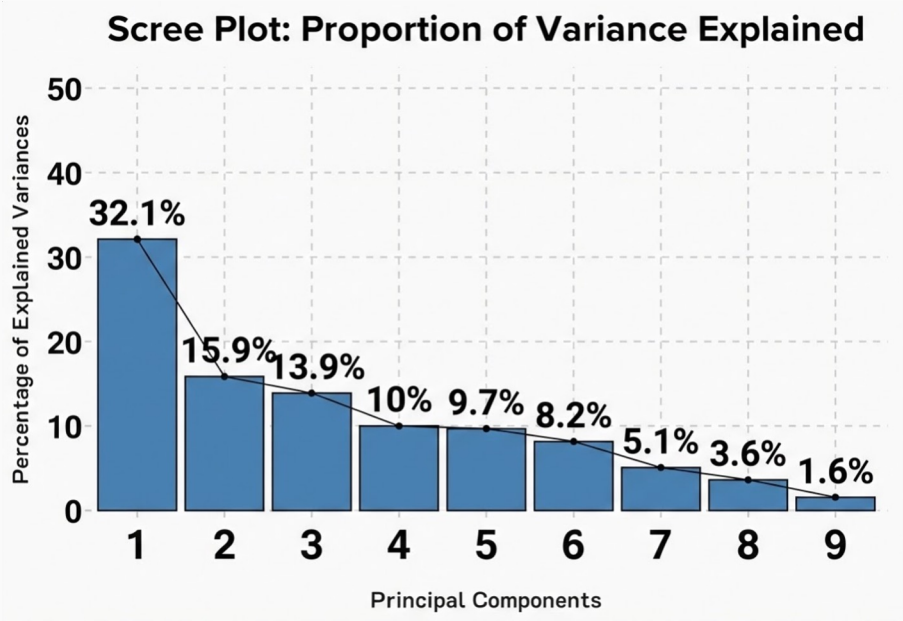
Standard Production rules & Market gaps



Negative Correlation
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

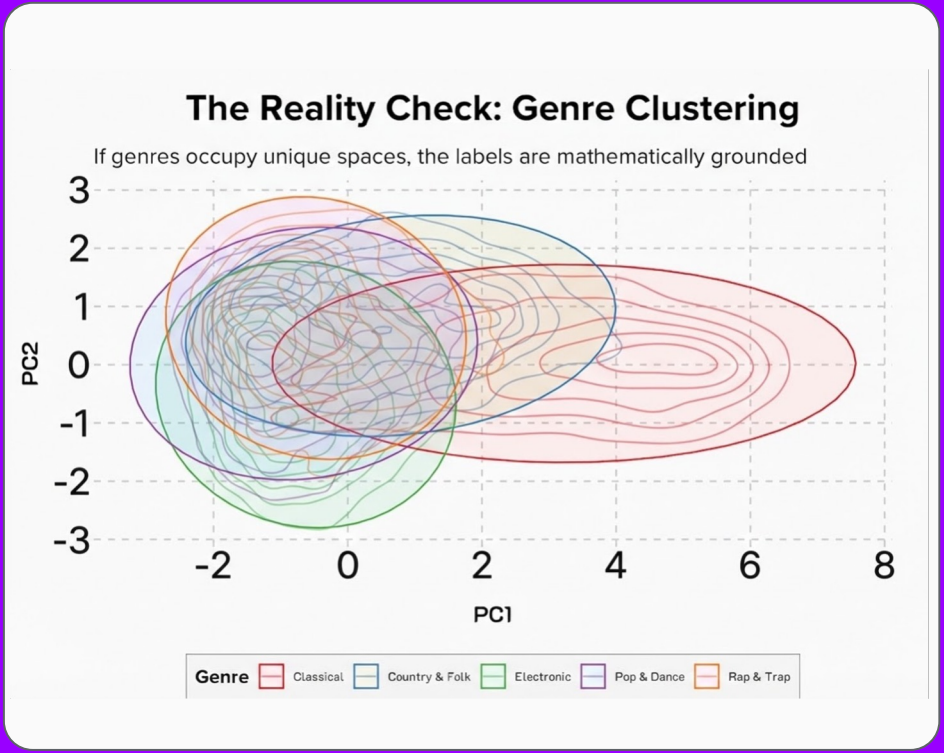
How algorithms cluster music



Scree PCA
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

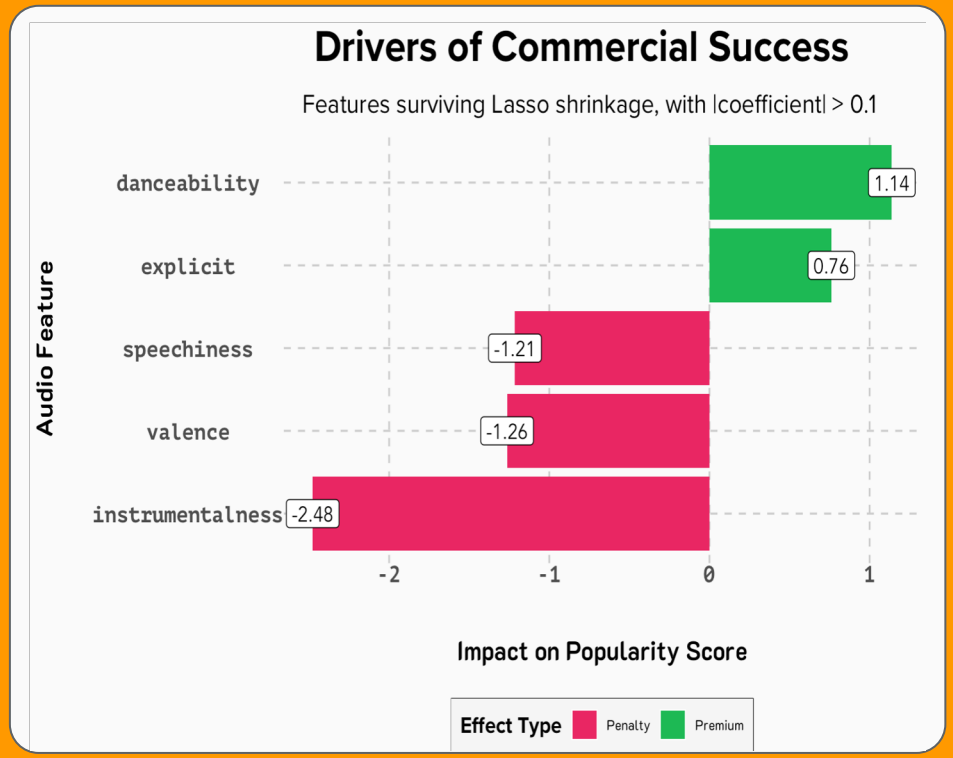
How algorithms cluster music



Genre Clustering
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

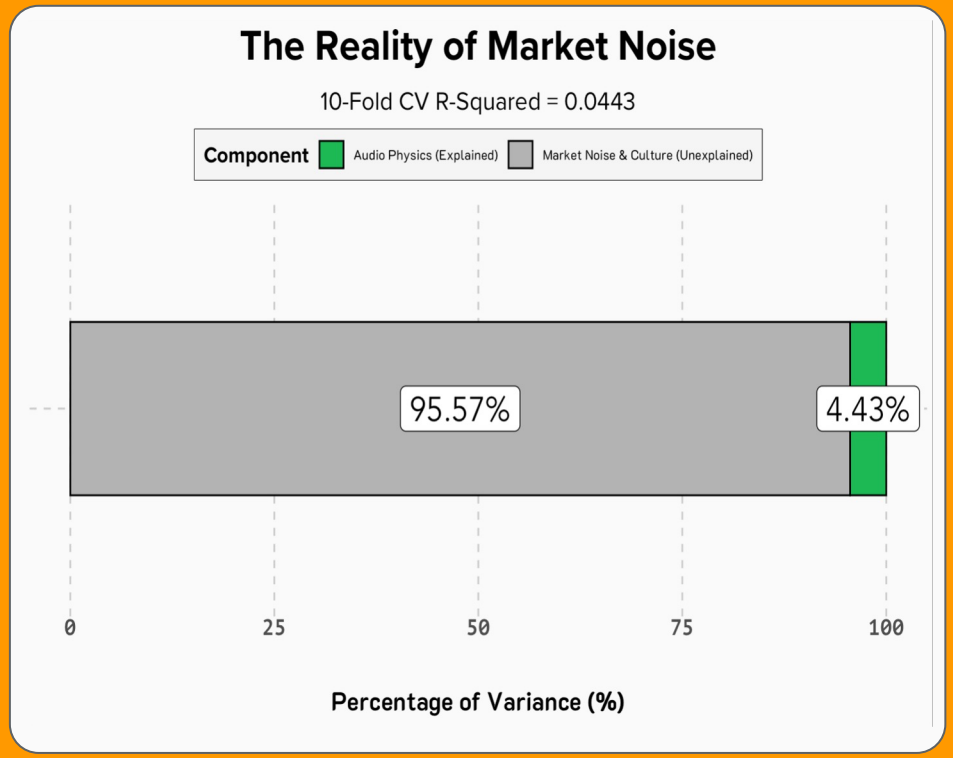
Predicting hit valuation through sonic math



Predicting Hits Through Improved Efficiency
Ft. RStudio
1:01
3:25

What do you want to analyse?
D&I

Predicting hit valuation through sonic math



Predicting Hits Through All Available Features
Ft. RStudio
1:01
3:25

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Future Considerations

Deploy unsupervised clustering (K-means) directly on the Principal Components → genres from scratch

Include exogenous data as Spotify streams, TikTok uses → have a comprehensive "cultural" view

Include time-series data → look at historical trends and changes within the industry

Thank You

The Winner Takes It All

ABBA

0:23 -3:25